

APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

FINAL ASSESSMENT EXAMINATION EXAMINATION • AY 2025-26

Course: Database Management Systems (CS501) | Faculty: Dr. Ravi Kumar

Date: 12 December 2025

Time: 120 Minutes

Max Marks: 40

Instructions: Answer all questions in Part A. Part B contains internal choice.

PART A — SQL & INDEXING OPERATIONS

Answer all questions. (16 Marks Total)

1. Write SQL queries to find the 2nd highest salary using dense_rank() window function. [4 M]
2. Construct a B+ Tree of order 3 inserting keys: 10, 20, 5, 15, 30, 25, 35. [6 M]
3. Explain how hashing collisions are handled in bucket overflow chains. [6 M]

PART B — TRANSACTION MANAGEMENT & CONCURRENCY CONTROL

Answer all questions. (24 Marks Total)

4. Prove why Strict Two-Phase Locking (Strict 2PL) guarantees conflict serializability and prevents cascading rollbacks. [12 M]
5. Describe the ARIES recovery algorithm phases: Analysis, Redo, and Undo with Write-Ahead Logging (WAL). [12 M]